EXERCISE 6

Variation of Constants: Five Particular Solutions

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Variation of Parameters · Euler Equation · Integral Answers

STATEMENT

Use the method of variation of constants to find a particular solution for the following equations:

- (i) $y'' - y' - 6y = e^x$;
- (ii) $\ddot{x} - x = t^{-1}$ (leave the answer as an integral);
- (iii) $y'' + 4y = \cot 2x$. Use that $\int \csc x \, dx = \ln|\csc x - \cot x|$;
- (iv) $t^2\ddot{x} - 2x = t^3$ (to find the solutions of the homogeneous equation, try solutions of the form $x = t^k$ with k a parameter);
- (v) $\ddot{x} - 4\dot{x} = \tan t$ (leave your answer as an integral).

PHASE 1: SETUP AND HYPOTHESES

Each equation is linear, second-order, inhomogeneous. Variation of constants (Lagrange; Robinson Ch. 18, Topic 3 Sec. 8) needs: (a) the equation in **standard form** $\ddot{x} + p\dot{x} + qx = g$ with the coefficient of the highest derivative equal to 1; (b) two linearly independent solutions x_1, x_2 of the homogeneous equation; (c) g continuous on an interval I where the coefficients are continuous. Then, on I,

$$x_p = -x_1 \int \frac{x_2\, g}{W}\, dt + x_2 \int \frac{x_1\, g}{W}\, dt, \qquad W = x_1\dot{x}_2 - x_2\dot{x}_1 \neq 0 \quad \text{(Robinson Eq. 18.8)}$$

Item	Standard form	Interval I where g and the coefficients are continuous
(i)	$y'' - y' - 6y = e^x$	$\mathbb{R}$
(ii)	$\ddot{x} - x = 1/t$	$t > 0$ (or $t < 0$)
(iii)	$y'' + 4y = \cot 2x$	$0 < x < \pi/2$ (between zeros of $\sin 2x$)
(iv)	$\ddot{x} - \frac{2}{t^2}x = t$ (divide by t^2)	$t > 0$ (or $t < 0$)

Item	Standard form	Interval I where g and the coefficients are continuous
(v)	$\ddot{x} - 4\dot{x} = \tan t$	$-\pi/2 < t < \pi/2$

PHASE 2: FORMULATION: THE DERIVATION OF THE FORMULA, RECALLED

Replace the constants of $x_h = c_1 x_1 + c_2 x_2$ by functions: $x = u_1 x_1 + u_2 x_2$. Choose $\dot{u}_1 x_1 + \dot{u}_2 x_2 = 0$ (18.7), so that $\dot{x} = u_1 \dot{x}_1 + u_2 \dot{x}_2$; substituting in the equation and using $L[x_1] = L[x_2] = 0$ leaves $\dot{u}_1 \dot{x}_1 + \dot{u}_2 \dot{x}_2 = g$. Cramer's rule on the system with determinant W gives

$$\dot{u}_1 = -\frac{x_2\, g}{W}, \qquad \dot{u}_2 = \frac{x_1\, g}{W}, \qquad x_p = u_1 x_1 + u_2 x_2$$

Constants of integration only add multiples of x_1, x_2, so they may be dropped.

METHOD OF THE CLASS NOTES [NOTES, P. 8]

This is the derivation of the notes: $x = u_1 x_1 + u_2 x_2$, impose $u_1' x_1 + u_2' x_2 = 0$, so $u_1' = -u_2' x_2 / x_1$ $(*)$; substituting in the equation leaves $u_1' x_1' + u_2' x_2' = f$, and with $(*)$ $u_2'(x_1 x_2' - x_2 x_1')/x_1 = f$, hence $u_2' = \frac{f\, x_1}{W}$ and $u_1' = -\frac{f\, x_2}{W}$. Cramer's rule gives the same without dividing by x_1.

Discrepancy with the notes: they print $u_1' = f\, x_2 / W$ without the minus sign, but $(*)$ gives $-f\, x_2 / W$. Check with $x'' + x' - 6x = 12$: the minus sign gives $x_N = -2$, which satisfies the equation; without it $x_N = -\frac{2}{5}$ and $L[x_N] = \frac{12}{5} \neq 12$.

PHASE 3: DERIVATION

(i) $y'' - y' - 6y = e^x$

Homogeneous equation: $y = e^{kx}$ gives $k^2 - k - 6 = (k-3)(k+2) = 0$, so $k = 3, -2$.

The homogeneous solutions are $y_1 = e^{3x}$ and $y_2 = e^{-2x}$. Their Wronskian is

$$W = y_1\, y_2' - y_2\, y_1' = (e^{3x})(-2e^{-2x}) - (e^{-2x})(3e^{3x}) = -5e^{x}$$

With $g = e^x$ (equation in standard form, coefficient of y'' equal to 1), Cramer's rule gives $u_1' = -\frac{y_2\, g}{W}$ and $u_2' = \frac{y_1\, g}{W}$:

$$u_1' = -\frac{(e^{-2x})(e^x)}{-5e^x} = \frac{e^{-2x}}{5}, \qquad u_2' = \frac{(e^{3x})(e^x)}{-5e^x} = -\frac{e^{3x}}{5}$$

Integrate (antiderivatives of exponentials, $\int e^{kx}dx = e^{kx}/k$):

$$u_1 = \int \frac{e^{-2x}}{5}\,dx = -\frac{e^{-2x}}{10}, \qquad u_2 = \int -\frac{e^{3x}}{5}\,dx = -\frac{e^{3x}}{15}$$

$$y_p = u_1 y_1 + u_2 y_2 = \left(-\frac{e^{-2x}}{10}\right)e^{3x} + \left(-\frac{e^{3x}}{15}\right)e^{-2x} = -\frac{e^x}{6} = -\frac{e^x}{6}$$

Check: $y_p' = -\frac{e^x}{6}$, $y_p'' = -\frac{e^x}{6}$, so $y_p'' - y_p' - 6y_p = (-\frac{e^x}{6}) - (-\frac{e^x}{6}) - 6(-\frac{e^x}{6}) = e^x$.

(ii) $\ddot{x} - x = t^{-1}$ (answer as an integral)

Homogeneous equation: $k^2 - 1 = 0$, so $k = \pm 1$: $x_1 = e^t$, $x_2 = e^{-t}$.

The homogeneous solutions are $x_1 = e^t$ and $x_2 = e^{-t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (e^t)(-e^{-t}) - (e^{-t})(e^t) = -2$$

With $g = \frac{1}{t}$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(e^{-t})(\frac{1}{t})}{-2} = \frac{e^{-t}}{2t}, \qquad u_2' = \frac{(e^{t})(\frac{1}{t})}{-2} = -\frac{e^{t}}{2t}$$

The antiderivatives of e^{-s}/s and e^s/s are not elementary (they are exponential integrals). We keep them as integrals from a fixed base point $t_0 > 0$ (the interval where $g = 1/t$ is continuous, $t > 0$), which is a legitimate antiderivative by the fundamental theorem of calculus:

$$u_1(t) = \frac{1}{2}\int_{t_0}^{t} \frac{e^{-s}}{s}\,ds, \qquad u_2(t) = -\frac{1}{2}\int_{t_0}^{t} \frac{e^{s}}{s}\,ds$$

$$\boxed{x_p(t) = \frac{e^t}{2}\int_{t_0}^{t} \frac{e^{-s}}{s}\,ds - \frac{e^{-t}}{2}\int_{t_0}^{t} \frac{e^{s}}{s}\,ds \quad (t, t_0 > 0)}$$

Combining the exponentials inside one integral, $x_p(t) = \dfrac{1}{2}\displaystyle\int_{t_0}^{t} \frac{e^{t-s} - e^{-(t-s)}}{s}\,ds = \int_{t_0}^{t} \frac{\sinh(t-s)}{s}\,ds$.

Verification by differentiation (Leibniz rule)

Let $I_1(t) = \int_{t_0}^{t} e^{-s}s^{-1}ds$ and $I_2(t) = \int_{t_0}^{t} e^{s}s^{-1}ds$, so $I_1' = e^{-t}/t$ and $I_2' = e^{t}/t$. Then $x_p = \frac{1}{2}(e^tI_1 - e^{-t}I_2)$ and, by the product rule,

$$\dot{x}_p = \frac{1}{2}\left(e^tI_1 + e^{-t}I_2\right) + \frac{1}{2}\left(e^t\frac{e^{-t}}{t} - e^{-t}\frac{e^t}{t}\right) = \frac{1}{2}\left(e^tI_1 + e^{-t}I_2\right)$$

$$\ddot{x}_p = \frac{1}{2}\left(e^tI_1 - e^{-t}I_2\right) + \frac{1}{2}\left(e^t\frac{e^{-t}}{t} + e^{-t}\frac{e^t}{t}\right) = x_p + \frac{1}{t}$$

so $\ddot{x}_p - x_p = 1/t$, as required.

(iii) $y'' + 4y = \cot 2x$

Homogeneous equation: $k^2 + 4 = 0$, $k = \pm 2i$, so $y_1 = \cos 2x$, $y_2 = \sin 2x$.

Wronskian (chain rule: $(\cos 2x)' = -2\sin 2x$, $(\sin 2x)' = 2\cos 2x$):

$$W = \cos 2x \cdot 2\cos 2x - \sin 2x \cdot (-2\sin 2x) = 2(\cos^2 2x + \sin^2 2x) = 2$$

Cramer's rule with $g = \cot 2x = \cos 2x / \sin 2x$:

$$u_1' = -\frac{\sin 2x\ \cot 2x}{2} = -\frac{\cos 2x}{2}, \qquad u_2' = \frac{\cos 2x\ \cot 2x}{2} = \frac{\cos^2 2x}{2\sin 2x}$$

For u_2' use $\cos^2 2x = 1 - \sin^2 2x$: $u_2' = \dfrac{1 - \sin^2 2x}{2\sin 2x} = \dfrac{1}{2}\left(\csc 2x - \sin 2x\right)$. Integrate with the substitution $s = 2x$, $ds = 2\,dx$, and the hint $\int \csc s\,ds = \ln\left|\csc s - \cot s\right|$:

$$u_1 = -\frac{1}{2}\int \cos 2x\,dx = -\frac{1}{2}\cdot\frac{\sin 2x}{2} = -\frac{\sin 2x}{4}$$

$$\int \csc 2x\,dx = \frac{1}{2}\int \csc s\,ds = \frac{1}{2}\ln|\csc 2x - \cot 2x|, \qquad \int \sin 2x\,dx = -\frac{\cos 2x}{2}$$

$$u_2 = \frac{1}{2}\left[\frac{1}{2}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{2}\right] = \frac{1}{4}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{4}$$

Assemble $y_p = u_1 y_1 + u_2 y_2$; the terms in $\sin 2x \cos 2x$ cancel:

$$y_p = -\frac{\sin 2x\cos 2x}{4} + \left[\frac{1}{4}\ln|\csc 2x - \cot 2x| + \frac{\cos 2x}{4}\right]\sin 2x = \frac{1}{4}\,\sin 2x\,\ln|\csc 2x - \cot 2x|$$

Since $\csc 2x - \cot 2x = \dfrac{1-\cos 2x}{\sin 2x} = \dfrac{2\sin^2 x}{2\sin x\cos x} = \tan x$, an equivalent form is $y_p = \frac{1}{4}\sin 2x\,\ln|\tan x|$.

Check by differentiating $y_p = \frac{1}{4}\sin 2x\,\ell$ with $\ell = \ln\tan x$, $\ell' = \frac{\sec^2 x}{\tan x} = \frac{2}{\sin 2x}$: $y_p' = \frac{1}{2}\cos 2x\,\ell + \frac{1}{4}\sin 2x\cdot\frac{2}{\sin 2x} = \frac{1}{2}\cos 2x\,\ell + \frac{1}{2}$; $y_p'' = -\sin 2x\,\ell + \frac{1}{2}\cos 2x\cdot\frac{2}{\sin 2x} = -\sin 2x\,\ell + \cot 2x$; hence $y_p'' + 4y_p = -\sin 2x\,\ell + \cot 2x + \sin 2x\,\ell = \cot 2x$.

(iv) $t^2\ddot{x} - 2x = t^3$

Homogeneous equation $t^2\ddot{x} - 2x = 0$. Try $x = t^k$ (chain/power rule): $\dot{x} = kt^{k-1}$, $\ddot{x} = k(k-1)t^{k-2}$, so $t^2\ddot{x} - 2x = [k(k-1) - 2]t^k = (k^2 - k - 2)t^k = (k-2)(k+1)t^k = 0$, hence $k = 2, -1$: $x_1 = t^2$, $x_2 = t^{-1}$ ($t > 0$).

Standard form: divide by t^2 (coefficient of $\ddot{x}$ must be 1): $\ddot{x} - \dfrac{2}{t^2}x = \dfrac{t^3}{t^2} = t$, so $g(t) = t$ (**not** t^3).

The homogeneous solutions are $x_1 = t^2$ and $x_2 = \frac{1}{t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (t^2)(-\frac{1}{t^2}) - (\frac{1}{t})(2t) = -3$$

With $g = t$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(\frac{1}{t})(t)}{-3} = \frac{1}{3}, \qquad u_2' = \frac{(t^2)(t)}{-3} = -\frac{t^3}{3}$$

$$u_1 = \int \frac{1}{3}\,dt = \frac{t}{3}, \qquad u_2 = \int -\frac{t^3}{3}\,dt = -\frac{t^4}{12}$$

$$x_p = u_1 x_1 + u_2 x_2 = \left(\frac{t}{3}\right) t^2 + \left(-\frac{t^4}{12}\right) t^{-1} = \frac{t^3}{4} = \frac{t^3}{4}$$

Check in the original equation: $\ddot{x}_p = \frac{3t}{2}$, so $t^2\ddot{x}_p - 2x_p = \frac{3t^3}{2} - \frac{t^3}{2} = t^3$.

(v) $\ddot{x} - 4\dot{x} = \tan t$ (answer as an integral)

Homogeneous equation: $k^2 - 4k = k(k-4) = 0, k = 0, 4$: $x_1 = 1, x_2 = e^{4t}$.

The homogeneous solutions are $x_1 = 1$ and $x_2 = e^{4t}$. Their Wronskian is

$$W = x_1\,x_2' - x_2\,x_1' = (1)(4e^{4t}) - (e^{4t})(0) = 4e^{4t}$$

With $g = \tan t$ (equation in standard form, coefficient of x'' equal to 1), Cramer's rule gives $u_1' = -\dfrac{x_2\,g}{W}$ and $u_2' = \dfrac{x_1\,g}{W}$:

$$u_1' = -\frac{(e^{4t})(\tan t)}{4e^{4t}} = -\frac{\tan t}{4}, \qquad u_2' = \frac{(1)(\tan t)}{4e^{4t}} = \frac{e^{-4t}\tan t}{4}$$

The first integral is elementary: $\int -\frac{1}{4}\tan t\,dt = \frac{1}{4}\ln|\cos t|$ (since $\frac{d}{dt}\ln\cos t = -\tan t$). The second, $\int e^{-4s}\tan s\,ds$, is not elementary, so it is left as an integral from a base point $t_0 \in (-\pi/2, \pi/2)$:

$$u_1(t) = \frac{1}{4}\ln\cos t, \qquad u_2(t) = \frac{1}{4}\int_{t_0}^{t} e^{-4s}\tan s\,ds$$

$$\boxed{x_p(t) = \frac{1}{4}\ln\cos t + \frac{e^{4t}}{4}\int_{t_0}^{t} e^{-4s}\tan s\,ds \qquad (|t|, |t_0| < \pi/2)}$$

Verification by differentiation

With $I(t) = \int_{t_0}^{t} e^{-4s} \tan s \, ds$, $I' = e^{-4t} \tan t$. Product rule: $\dot{x}_p = -\frac{1}{4} \tan t + e^{4t} I + \frac{e^{4t}}{4} e^{-4t} \tan t = e^{4t} I$ (the two tangent terms cancel). Again, $\ddot{x}_p = 4e^{4t} I + e^{4t} e^{-4t} \tan t = 4\dot{x}_p + \tan t$, i.e. $\ddot{x}_p - 4\dot{x}_p = \tan t$.

PHASE 4: INTERPRETATION AND CHECKS

Item	Particular solution x_p	Domain
(i)	$-\frac{e^x}{6}$	$\mathbb{R}$
(ii)	$\frac{e^t}{2} \int_{t_0}^{t} \frac{e^{-s}}{s} ds - \frac{e^{-t}}{2} \int_{t_0}^{t} \frac{e^s}{s} ds$	$t > 0$
(iii)	$\frac{1}{4} \sin 2x \, \ln \lvert \csc 2x - \cot 2x \rvert = \frac{1}{4} \sin 2x \, \ln \lvert \tan x \rvert$	$0 < x < \pi/2$
(iv)	$\frac{t^3}{4}$	$t > 0$
(v)	$\frac{1}{4} \ln \cos t + \frac{e^{4t}}{4} \int_{t_0}^{t} e^{-4s} \tan s \, ds$	$\lvert t \rvert < \pi/2$

- The general solution of each equation is $x_h + x_p$ with the two homogeneous solutions used above.
- (i) and (iv) could also have been found by undetermined coefficients (Ke^x and Kt^3); both methods agree ($-\frac{1}{6}e^x$ and $\frac{1}{4}t^3$), a useful cross-check of the Wronskian bookkeeping. (iii) and (v) have forcings (cot, tan) outside the family of undetermined coefficients, which is the reason for the method.
- (iv) is an Euler equation: the substitution $x = t^k$ turns it into an algebraic equation for k, and its standard form needs the division by t^2 before applying the formula.
- Wronskians: $-5e^x$ in (i) (consistent with Abel's identity, $W = W_0 e^x$ since $p = -1$), -2 in (ii), 2 in (iii), -3 in (iv) and $4e^{4t}$ in (v) (Abel: $p = -4$, $W = W_0 e^{4t}$); none vanishes on its interval.
- The sheet works in dimensionless variables, so no unit check is attached to the forcings (t^3, e^x, $\cot 2x$, $\tan t$ are pure numbers); the check that matters is that each particular solution reproduces its forcing exactly, which was done symbolically.
- Every particular solution (and the integral forms, via the differentiation above) was verified symbolically in `scripts/verify_topic3_topic4.py`.